University Digital Library Management System

Owner:
Reviewer:
Contributors:
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Executive Summary

High level system description

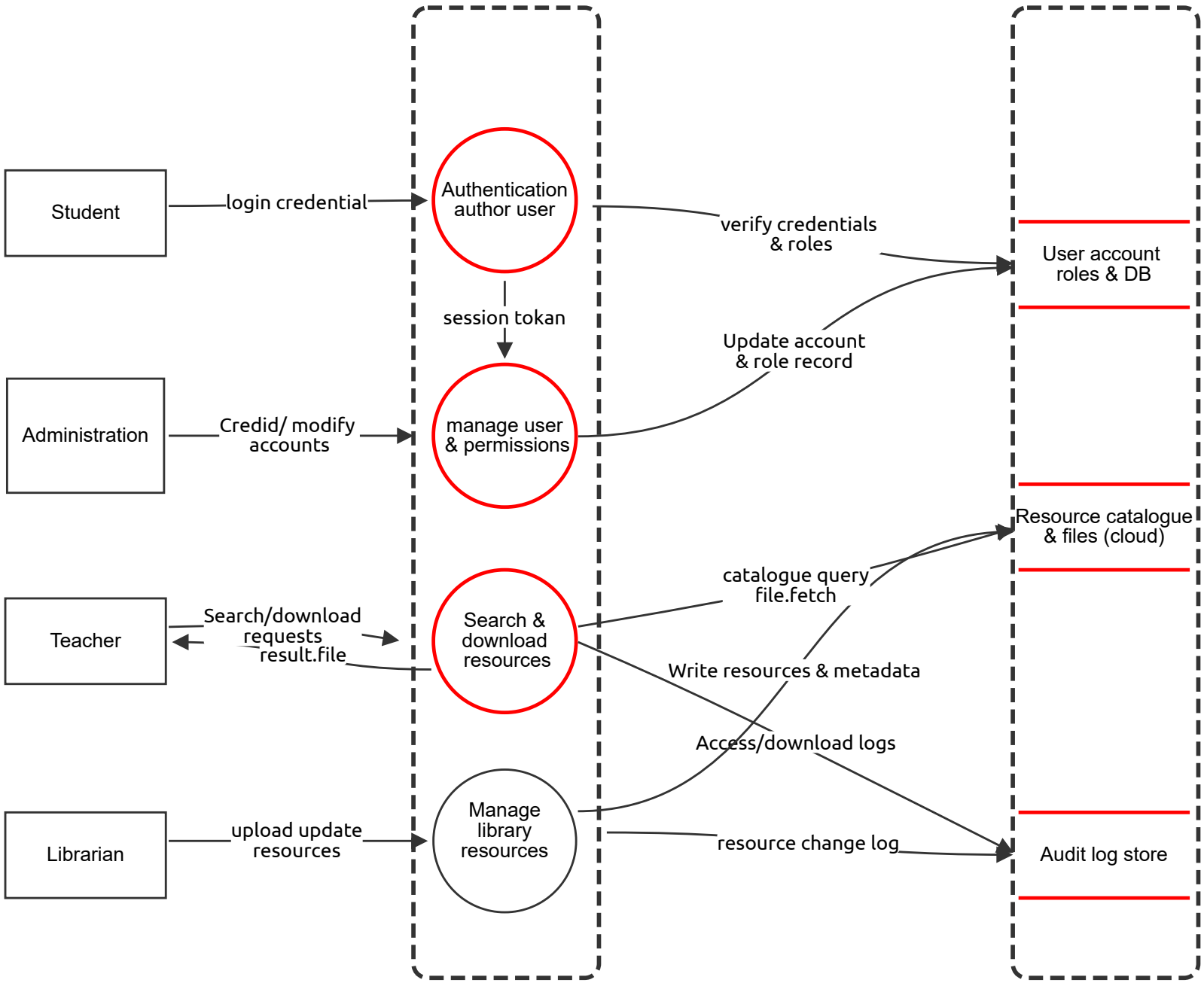
Not provided

Summary

Total Threats	8
Total Mitigated	0
Total Open	8
Open / Critical Severity	0
Open / High Severity	6
Open / Medium Severity	2
Open / Low Severity	0

New STRIDE diagramUniversity Digital Library

Data flow diagram for the University Digital Library Management System



New STRIDE diagramUniversity Digital Library

Student (Actor)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Administration (Actor)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Librarian (Actor)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Search & download resources (Process)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
5	Search & Download Resources / Cloud Files	Denial of service	High	Open		An attacker may send a large number of requests to the library system, consuming server resources and preventing legitimate students and teachers from accessing	Apply rate limiting, traffic filtering, resource monitoring, and suitable availability protection.

Authentication author user (Process)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
1	Unauthorized Login	Spoofing	High	Open		An attacker may steal a student's or teacher's password and log in using their account without permission. This could expose private account information or allow unauthorized actions.	Use multi-factor authentication (MFA), strong passwords, secure password hashing, and login rate limiting.
8	Session Token Theft	Spoofing	High	Open		An attacker may steal a valid session token and use it to impersonate a logged-in user without knowing the user's password.	Use HTTPS, secure session cookies, appropriate token expiration, and session invalidation after logout.

manage user & permissions (Process)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
2	Manage User & Permissions	Elevation of privilege	High	Open		A student may exploit weak access controls to gain librarian or administrator privileges and modify accounts or library resources.	Implement Role-Based Access Control (RBAC), enforce least privilege, and verify permissions on the server for every sensitive action.

Manage library resources (Process)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Teacher (Actor)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

User account roles & DB (Store)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
3	User Accounts & Roles DB / Resource Catalogue & Files	Tampering	High	Open		An attacker may modify user roles, book records, or digital files without authorization, affecting the integrity of library data.	Enforce database access controls, validate changes, restrict write permissions, and maintain secure backups.

login credential (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

verify credentials & roles (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

session tokan (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Update account & role record (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

result.file (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Access/download logs (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

resource change logs (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Credid/ modify accounts (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Search/download requests (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

upload update resources (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Write resources & metadata (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

catalogue query file.fetch (Data Flow)

Number	Title	Type	Severity	Status	Score	Description	Mitigations

Audit log store (Store)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
6	Audit Log Store	Tampering	Medium	Open		An attacker or unauthorized user may delete or modify audit logs to hide suspicious activities, making security investigations difficult.	Restrict log access, use centralized tamper-resistant logging, and maintain protected log backups.
7	Repudiation of Actions	Repudiation	Medium	Open		A librarian may deny uploading, deleting, or modifying a resource if the system does not maintain reliable records of who performed the action and when.	Record authenticated user IDs, timestamps, and resource changes in protected audit logs.

Resource catalogue & files (cloud) (Store)

Number	Title	Type	Severity	Status	Score	Description	Mitigations
4	Search & Download Resources / Cloud Files	Information disclosure	High	Open		A user may access restricted research papers, private account details, or other protected information because of incorrect permissions or insecure cloud storage settings.	Enforce authorization checks, encrypt sensitive data in transit and at rest, and configure cloud storage permissions securely.